WORKSHEET 1.6

Trapping a Limit: The Squeeze Theorem

Unit 1: Limits and Continuity

Key ideas

- **Squeeze theorem.** If $g(x) \le f(x) \le h(x)$ near c and $\lim_{x\to c} g(x) = \lim_{x\to c} h(x) = L$, then $\lim_{x\to c} f(x) = L$.
- Bounds usually come from $-1 \le \sin(\cdot) \le 1$ or $-1 \le \cos(\cdot) \le 1$, multiplied by a non-negative factor.
- $\lim_{\theta\to 0} \dfrac{\sin\theta}{\theta} = 1$ and $\lim_{\theta\to 0} \dfrac{1-\cos\theta}{\theta} = 0$, with θ in radians.
- The expression inside the sine must match the expression underneath it.

Worked examples

Example 1 · Matching the argument

Evaluate $\lim_{x\to 0} \dfrac{\sin(7x)}{2x}$.

Solution. $\dfrac{\sin(7x)}{2x} = \dfrac{7}{2} \cdot \dfrac{\sin(7x)}{7x} \to \dfrac{7}{2} \cdot 1.$

The limit is $\dfrac{7}{2}$.

Example 2 · A function known only by its bounds

A function satisfies $2x + 1 \le f(x) \le x^2 + 2$ for all x. Find $\lim_{x\to 1} f(x)$.

Solution. $\lim_{x\to 1}(2x+1) = 3$ and $\lim_{x\to 1}(x^2+2) = 3$.

By the squeeze theorem, $\lim_{x\to 1} f(x) = 3$.

Example 3 · A full squeeze argument

Show that $\lim_{x\to 0} x^2 \sin\left(\dfrac{3}{x}\right) = 0$.

Solution. For $x \neq 0$, $-1 \le \sin\left(\frac{3}{x}\right) \le 1$. Since $x^2 \ge 0$, $-x^2 \le x^2 \sin\left(\frac{3}{x}\right) \le x^2$. Also $\lim_{x\to 0}(-x^2) = 0$ and $\lim_{x\to 0} x^2 = 0$.

By the squeeze theorem the limit is 0.

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Practice

CORE

1. Evaluate $\lim_{x\to 0} \dfrac{\sin(4x)}{x}$.

2. Evaluate $\lim_{x\to 0} \dfrac{\sin x}{5x}$.

3. Evaluate $\lim_{x\to 0} \dfrac{\tan(3x)}{x}$.

4. Evaluate $\lim_{x\to 0} \dfrac{x}{\sin(2x)}$.

EXAM LEVEL

5. Evaluate $\lim_{x\to 0} \dfrac{\sin(6x)}{\sin(2x)}$.

6. Evaluate $\lim_{x\to 0} \dfrac{1-\cos(2x)}{x^2}$.

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7. A function satisfies $5 - x^2 \le g(x) \le 5 + x^4$ for all x. Find $\lim_{x \to 0} g(x)$, with justification.

8. Evaluate $\lim_{x \to \infty} \dfrac{3 + \sin x}{x}$, with justification.

Question 9 · Free-response practice

Let $f(x) = x^3 \cos\left(\dfrac{1}{x}\right)$ for $x \neq 0$.

(a) Explain why $\lim_{x \to 0} f(x)$ cannot be found with the product law.

(b) Use the squeeze theorem to find $\lim_{x \to 0} f(x)$. (Take care: x^3 can be negative.)

(c) Does the same method show that $\lim_{x \to 0} (x + 1) \cos\left(\dfrac{1}{x}\right)$ exists? Explain.

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Worked solutions

1. 4.
2. $\frac{1}{5}$.
3. $\frac{\sin(3x)}{x} \cdot \frac{1}{\cos(3x)} \to 3$.
4. $\frac{1}{2} \cdot \frac{2x}{\sin(2x)} \to \frac{1}{2}$.
5. $\frac{\sin 6x}{6x} \cdot \frac{2x}{\sin 2x} \cdot \frac{6}{2} \to 3$.
6. $\frac{1-\cos 2x}{x^2} \cdot \frac{1+\cos 2x}{1+\cos 2x} = \frac{\sin^2(2x)}{x^2} \cdot \frac{1}{1+\cos 2x} \to 4 \cdot \frac{1}{2} = 2$.
7. $\lim_{x\to 0}(5-x^2) = 5$ and $\lim_{x\to 0}(5+x^4) = 5$, so by the squeeze theorem the limit is 5.
8. For $x > 0$, $\frac{2}{x} \le \frac{3+\sin x}{x} \le \frac{4}{x}$. Both bounds tend to 0 as $x \to \infty$, so by the squeeze theorem the limit is 0.
9. (a) $\lim_{x\to 0} \cos\left(\frac{1}{x}\right)$ does not exist (the values oscillate), and the product law needs both limits to exist.
   (b) $\left|\cos\frac{1}{x}\right| \le 1$, so $-|x|^3 \le x^3 \cos\frac{1}{x} \le |x|^3$. Both bounds tend to 0 as $x \to 0$, so by the squeeze theorem the limit is 0.
   (c) No. The bounds would be $-|x+1|$ and $|x+1|$, which tend to -1 and 1: not equal. In fact the values oscillate between about -1 and 1, and the limit does not exist.

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